

Editor's Choice

Indentation and scratching of iron by a rotating tool – a molecular dynamics study

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ARTICLE INFO

Keywords:

Molecular dynamics
Nanoindentation
Nanomachining
Dislocations
Plasticity
Grinding
Polishing
Drilling

ABSTRACT

When abrasive particles interact with a surface, besides translational center-of-mass motion also rotational motion of the abrasive may be involved. We perform molecular dynamics simulations of the influence of the rotational motion in a model scenario, where both translational and rotational velocities are controlled. We find that for angular velocities corresponding to slip-free rolling, the influence of rotation is small and shows up as a decrease of hardness and of friction. The effect is somewhat more pronounced for rolling motion (rotation axis parallel to surface) than for drilling motion (rotation axis perpendicular to surface). Pile-up structures are unaffected. Only if rotational speeds strongly exceed the rolling-without-slip limit, strong effects of the rotational speed are observed: the machined material amorphizes rather than to develop dislocation plasticity, the groove develops oscillatory patterns, and the material hardness is strongly reduced.

1. Introduction

Nanoindentation [1–13] and scratching [14–24] of materials have been studied in great detail using molecular dynamics simulation. These techniques have traditionally been used as a first approach to the study of nanomachining/micromachining processes. However, in several applications of surface treatment of materials, besides translational motion also rotational motion is involved. An example is provided by surface polishing using hard abrasive particles or by grinding; there small particles move over the surface and have both translational and rotational degrees of motion. Surface polishing and grinding are categorized as mechanical micromachining processes. Their application spans a wide range of materials including metals, semiconductors, ceramics, glasses and polymers for a variety of applications, from optics to electronics.

The mechanisms of surface machining combining translational and rotational motion have been studied comparatively less. Si et al. [25] studied rolling motion of a silica particle on top of a silicon surface and demonstrated the effects on material removal from the surface. Yang et al. [26] investigated the so-called ultra-precision mechanical polishing process of Si by diamond spheres of 4 nm radius and found that only small rotation velocities lead to good surface qualities. For a Cu sample,

these authors found that with increasing rotation speed, the machining mechanism changes from a cutting to a ploughing regime in that the chip formed in front of the scratch groove disappears at higher rotation velocity [27]. The effects of a rolling abrasive on a Cu surface [28] and of surface texture on grinding of a Cu surface were studied by Li et al. [29]. Wu et al. [30,31] studied nanodrilling of Cu and concluded that higher rotation velocities render the machining process more efficient, but also lead to the creation of an amorphous zone. More recently, in a series of papers [32–34], Nguyen and Fang studied polishing of Si and SiC samples by hard diamond grains with radii of 1–2 nm and concluded that the rolling motion aids in the surface removal process. We note that the technique of vibration-assisted machining bears some similarities to these studies, as it combines translational motion and elliptical movement of the tool; a recent study on pure iron [35] reported a significant reduction in forces as well as significant changes in the deformation mechanisms. Also the technique of rotary ultrasonic-assisted machining combines translational and rotational motion of the tool and is used to machine hard and brittle materials [36].

Here we present a model study, in which a spherical grain interacts with a Fe surface. As the effect of the translational motion has been repeatedly studied, we fix its velocity to 20 m/s. We study the effect of

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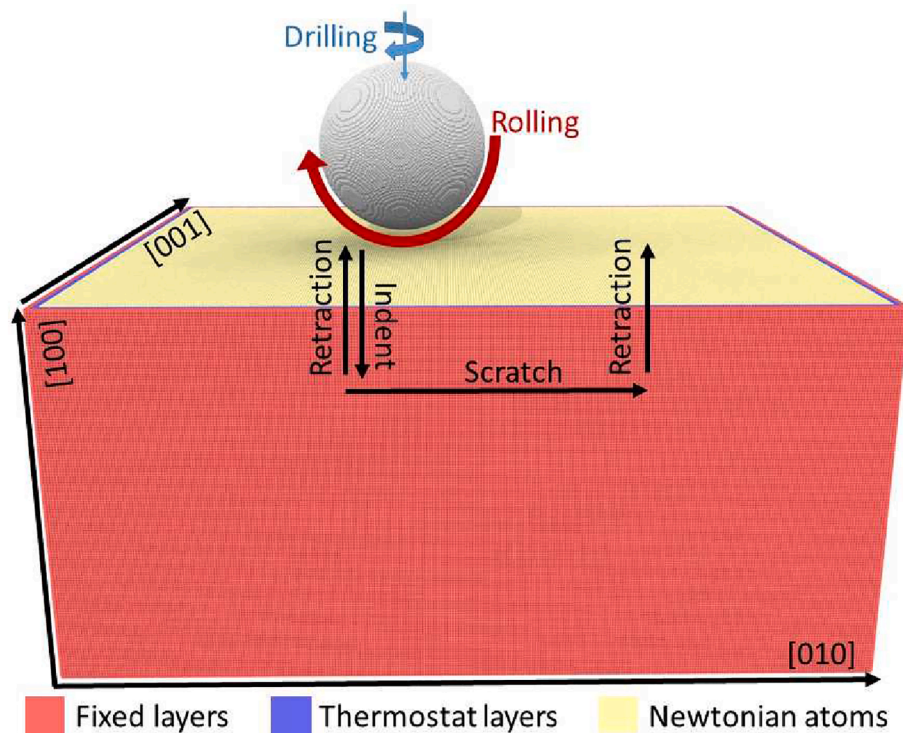


Fig. 1. Schematics of the simulation setup.

rotation by varying the angular velocity in a wide range, by several orders of magnitude. Also the effect of the orientation of the rotation axis is studied: both ‘drilling’ – with the rotation axis perpendicular to the surface – and ‘rolling’ are investigated. In addition, we consider the processes of indentation, where the particle is pushed perpendicularly into the sample, and of scratching, where it moves parallel to the surface. Thus a wide range of scenarios is investigated, which allow us to draw conclusions on the circumstances where rotational motion considerably influences the machining process.

2. Method

The sample is a bcc Fe crystal with a (100) surface; its sides are aligned with the [010] and [001] crystal axes. The crystal has lateral sizes of 74.8 nm $\times$ 85.0 nm and a depth of 41 nm, see Fig. 1, and consists in total of 22,407,812 atoms. The spherical tool is built of a rigid array of 738,612 carbon atoms in diamond structure; it has a radius of $R = 10$ nm.

The interaction between Fe atoms is described by the Mendelev potential [37]. Two atomic layers of the substrate at the bottom and the lateral sides have been fixed to prevent the whole substrate from translational or rotational movement. The next four layers are kept at a temperature of <10 mK by a Langevin thermostat. Fe and C atoms interact purely repulsively; the interaction potential is obtained from a Lennard-Jones potential [38] that has been cut off at its minimum at 4.2 Å [39] and is then shifted such as to have continuous energy and force at the cut-off radius.

In this study, two kinds of simulations are performed. The first is an indentation simulation with results discussed in Section 3. Here, the tool is initially positioned 5 Å above the surface; at time $t = 0$ it starts moving perpendicular into the sample with a velocity of $v = 20$ m/s. We penetrate to a final depth of $d = 4$ nm, which is reached at $t = 225$ ps; after a holding phase of 100 ps, the tool is retracted again with velocity 20 m/s.

The second kind of simulation is called scratching simulation and discussed in Section 4. It starts after the indentation and holding stage

with the tool indented to a depth of 4 nm. Then the tool moves along the [010] crystal direction at constant depth with velocity 20 m/s for a scratching length of 15 nm. After a holding phase of 100 ps, the tool is retracted with a velocity of 20 m/s. The indentation, scratching and retraction stages of the tool motion are illustrated schematically in Fig. 1.

While the translational velocity of the tool has been fixed in all stages to 20 m/s, we consider different rotational velocities in this study. The tool rotates both during the indentation and the scratching stages. We denote motion with the rotation axis perpendicular to the surface as ‘drilling’, and motion with the rotation axis parallel to the surface plane but perpendicular to the scratching direction as ‘rolling’, see Fig. 1. In each case we study several angular velocities ω : 0, 0.02, 0.36, 3.6 and 36 °/ps. We note that the angular velocity corresponding to slip-free rolling of the tool on the surface corresponds to $\omega_{sf} = v/R = 0.115$ °/ps; we thus study both angular velocities below and above this slip-free rolling case. Only rolling motion with a rotation direction that pushes material to the back is considered, see Fig. 1, as it could arise from an abrasive particle rolling over the surface. We present and discuss the major findings in the main text; additional data are found in the Supplementary Material (SM) [40].

We use the open-source LAMMPS code [41] with a constant time step of 1 fs to perform the simulations. The use of a tool that performs both translation and rotation is not straightforward to introduce in LAMMPS, except during the drilling mode of indentation where the rotation axis is aligned with the translation motion. LAMMPS does not provide commands to allow the rotation axis to move in space. In order to circumvent this problem, we fix the tool at a specific position while the entire sample is moved at constant translational velocity. We demonstrate in Appendix A the validity of this procedure.

The dislocation extraction algorithm (DXA) [42] is used to identify the dislocations, to determine their Burgers vectors $\mathbf{b}$, and to measure the total length of the dislocation lines, L_{disl} . The free software tools ParaView [43] and OVITO [44] are employed to visualize the atomistic configurations.

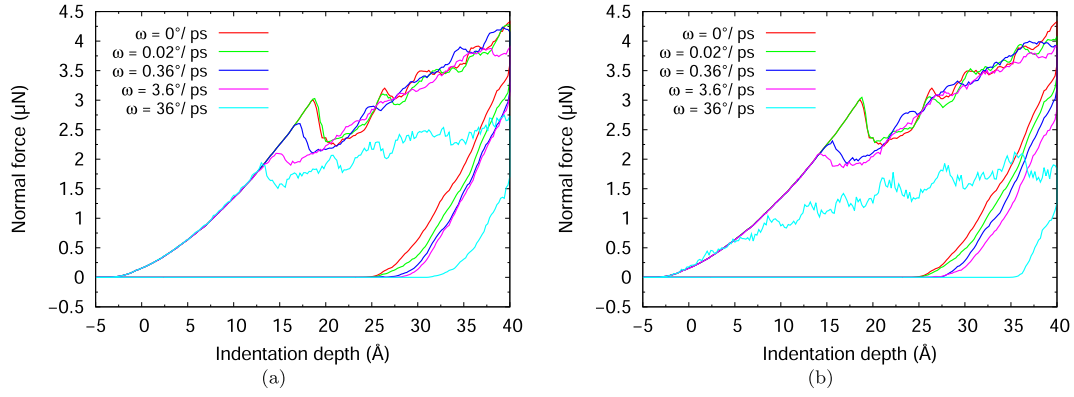


Fig. 2. Evolution of the normal force with indentation depth for (a) drilling and (b) rolling.

Table 1

Indentation hardness H averaged over the last 20 Å of indentation and characteristics of the plastic zone after full indentation. Data are for various angular rotation velocities, ω . L_{dist} : total dislocation length within plastic zone. R_{pl} : size of plastic zone. f : plastic-zone size factor. ρ : dislocation density.

	Drilling					Rolling				
ω (°/ps)	0	0.02	0.36	3.6	36	0.02	0.36	3.6	36	
H (GPa)	20.0	19.8	19.9	19.4	10.8	19.6	19.9	19.0	7.4	
L_{dist} (Å)	8908	8865	10112	9670	5106	8538	10697	9917	65	
R_{pl} (Å)	286	380	320	304	311	327	356	292	140	
f	3.6	4.8	4.0	3.8	3.9	4.1	4.5	3.7	1.8	
ρ (10^{16} m^{-2})	1.82	0.78	1.47	1.64	0.82	1.17	1.13	1.90	0.12	

3. Indentation

3.1. Forces

In Fig. 2, the variation of the force exerted on the sample during indentation is shown. The data show the generic behavior of (i) an elastic increase, (ii) load drop after nucleation of dislocations in the sample, followed by (iii) a further force increase in the plastic stage. After the maximum indentation depth of 4 nm is reached, the force (iv) drops during the holding stage, and then (vi) smoothly decreases to zero as the tool is retracted from the sample.

For moderate rotation velocities, up to around the slip-free rolling velocity ω_{sf} and even for $\omega = 3.6^\circ/\text{ps}$, the influence of the rotation is minor. It is only felt in the onset of plasticity, which occurs earlier for increasing ω , both for the cases of drilling and rolling rotation. The

larger disturbance of the material by the tool rotation alleviates dislocation nucleation in the sample, as the tool rotation imposes shear stress components on the surface. The influence of tool rotation is also felt during tool retraction in that the tool loses contact with the sample earlier for higher ω .

We tabulate in Table 1 the contact pressure averaged over the last 20 Å of the indentation process, corresponding to the plastic stage of indentation. The contact pressure is calculated from the ratio of the normal force and contact area of the tool projected into the surface plane; its average represents the material hardness. The data in Table 1 demonstrate a slight tendency of decreasing hardness with increasing rotation velocity, which becomes particularly pronounced for $\omega = 3.6^\circ/\text{ps}$; but even then, it amounts to only a 5% reduction.

The data for the highest rotation velocity, $\omega = 36^\circ/\text{ps}$, strongly deviates from the other force-depth curves. The plastic regime is entered

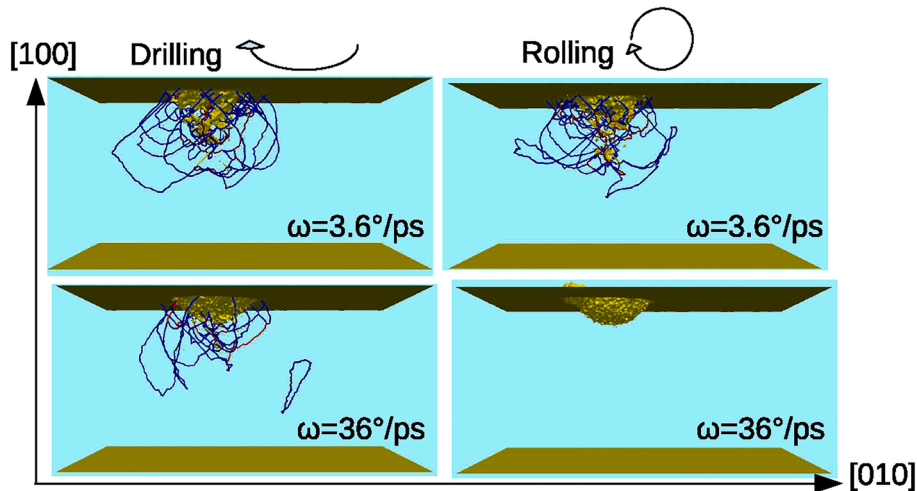


Fig. 3. Snapshots of the defects formed after the indentation stage. Yellow: deformed surface and other unidentified defects. Dislocations are colored according to their Burgers vector b : blue $\frac{1}{2}\langle 111 \rangle$, red $\langle 100 \rangle$.

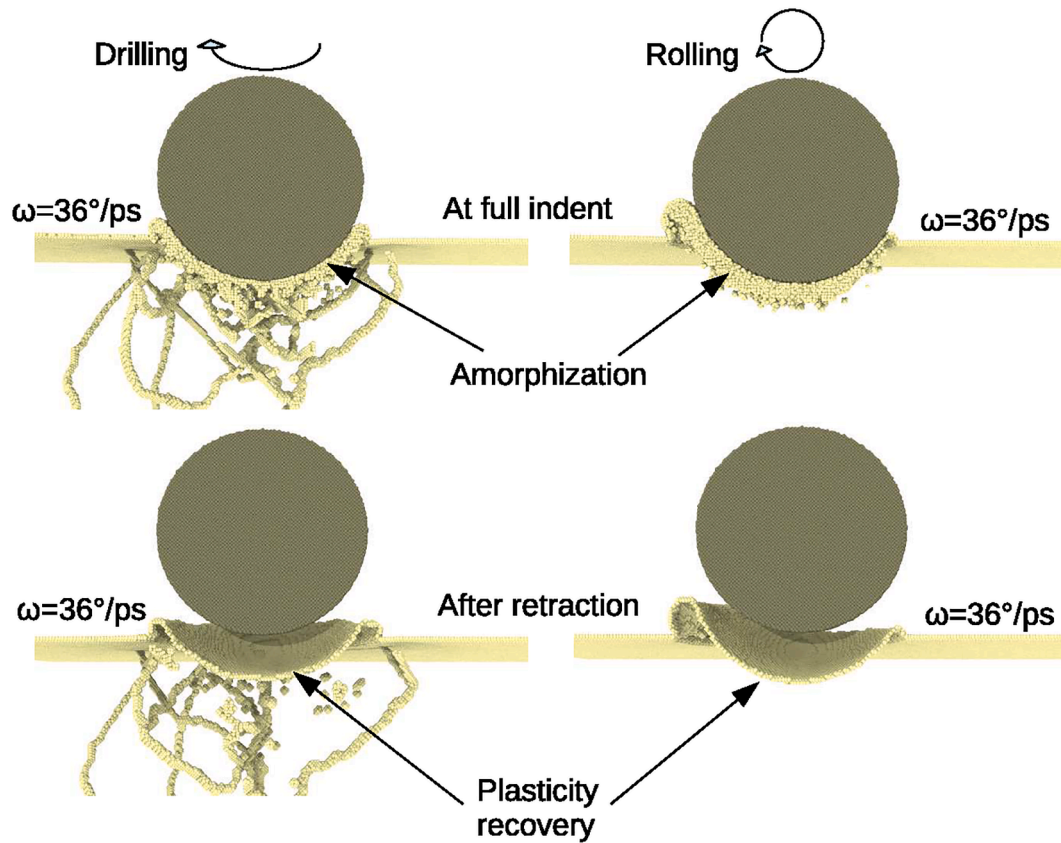


Fig. 4. Snapshots of the defects formed at the end of the indentation stage after full indentation (top) and after tool retraction (bottom). Data are shown both for drilling (left) and for rolling (right), for an angular velocity of $\omega = 36^\circ/\text{ps}$.

most quickly, also the force during the plastic regime is considerably below that of the other cases of smaller rotation velocity, and the tool loses contact with the indent pit most quickly. The hardness for this high value of the rotation speed is decreased by around 50% for drilling motion and even more in the case of rolling.

3.2. Dislocations and indent pit

The dislocations created during indentation are displayed in Fig. 3. Here only the two cases of largest rotation velocity are shown; the other cases are presented in the SM [40], as they show no discernible differences to the case of $\omega = 3.6^\circ/\text{ps}$ depicted here. The plastic zone is filled with dislocations of Burgers vectors $\mathbf{b} = \frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$, as they are typical for indentation into bcc crystals [22,12]. These dislocations form shear loops as they expand from their nucleation site beneath the tool. For the largest rotation velocity, $\omega = 36^\circ/\text{ps}$, dislocation formation is reduced and in the case of the rolling mode almost entirely stopped. These findings are quantified in Table 1, where the total dislocation length, L_{disl} , is tabulated. Only little variation of L_{disl} with ω – below around 20% compared to the non-rotating case – is observed. Both for the drilling and the rolling mode, a trend to a slightly increased dislocation production with increasing rotation might be read off the data; this is in line with the slightly reduced material hardness found. For the largest rotation velocity, $\omega = 36^\circ/\text{ps}$, a clear decrease of L_{disl} is found.

The peculiarities of the case of the highest rotation velocity, $\omega = 36^\circ/\text{ps}$, are further illustrated in Fig. 4, where the indent pit created during indentation is shown both for drilling and rolling motion. In both cases, at deepest indentation, a considerable pileup has formed, which is asymmetrical for the rolling mode in that material has been moved towards the rear side of the tool. The pileup and also the entire zone adjacent to the tool has amorphized; this explains why less dislocations

have formed at this rotation speed: damage creation proceeds via transformation to the amorphous phase rather than by dislocation formation. After retraction of the tool, the amorphous material recrystallizes. Due to the creation of an amorphous contact zone, the tool loses contact with the sample quite early, since elastic rebound of the material – which happens in the case of lower rotation velocity – is suppressed in this case.

For the cases of smaller rotation velocity, the indent pit always has the shape of a spherical cap as the image of the indenting sphere. No pileup is formed and no influence of drilling or rolling mode of rotation is observed, see SM [40].

Table 1 displays several quantities characterizing the plasticity created in the indent. The radius of the plastic zone, R_{pl} , is determined as the largest distance to the dislocations that still adhere to the imprint surface from the center of the tool [45]. By scaling this quantity to the radius of the contact zone, $a_c = (R^2 - (R - d)^2)^{1/2} = 8 \text{ nm}$, we obtain the so-called plastic-zone size factor, $f = R_{\text{pl}}/a_c$ [46]. Finally, we determine the average dislocation density in the plastic volume as the total dislocation length, L_{disl} , divided by the volume of the plastic zone, which we approximate as a hemisphere of radius R_{pl} . The data assembled in Table 1 show no systematic change with rotation velocity; values of $f = 3\text{--}5$ as well as the fluctuations showing up in R_{pl} (and f) are typical of indentation simulations and are caused by the randomness in dislocation nucleation and in the expansion of the dislocation network [12]. Also values of the dislocation density in the plastic zone of around $1 \times 10^{16} \text{ m}^{-2}$ are standard. The only deviation is seen for the rolling simulation with the largest angular velocity, $\omega = 36^\circ/\text{ps}$, where both the size of the plastic zone is extremely small, $f = 1.8$, and the dislocation density is one order of magnitude below that of the other simulations. As discussed above, this is caused by the amorphization of the crystal in the vicinity of the swiftly rotating tool, which prevents dislocations from

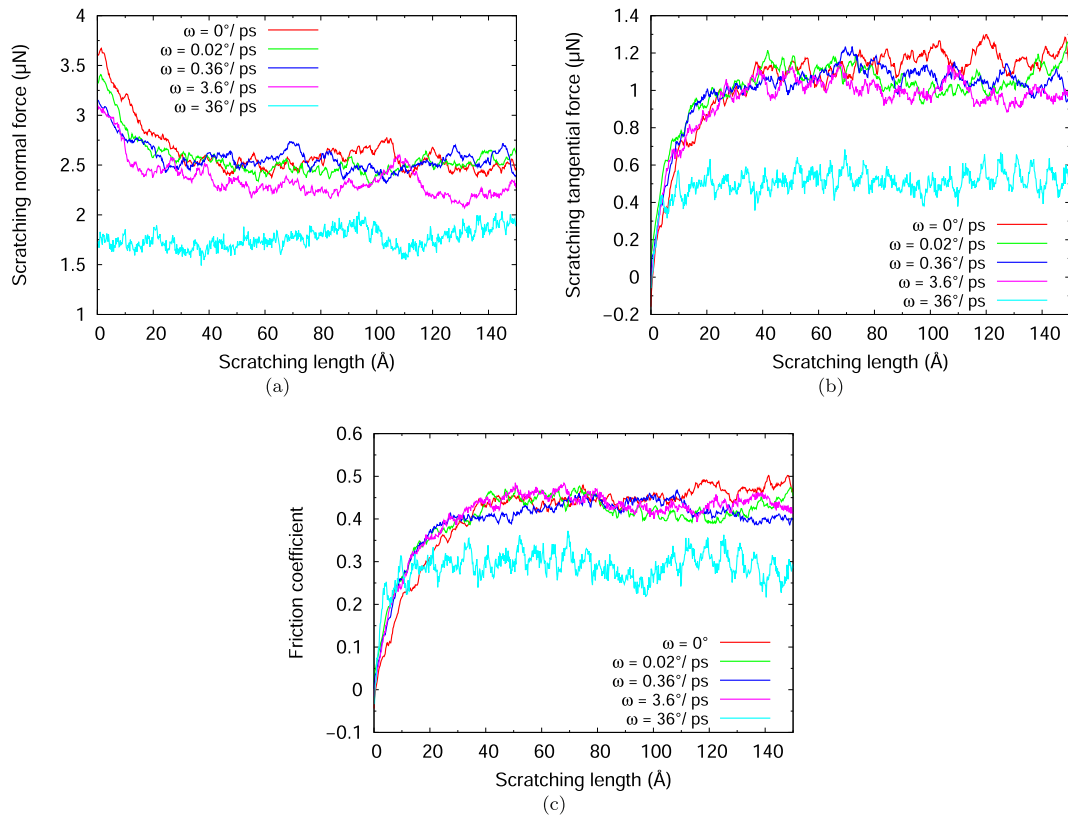


Fig. 5. Evolution of the (a) normal and (b) tangential force and of the (c) friction coefficient with scratching length for drilling motion.

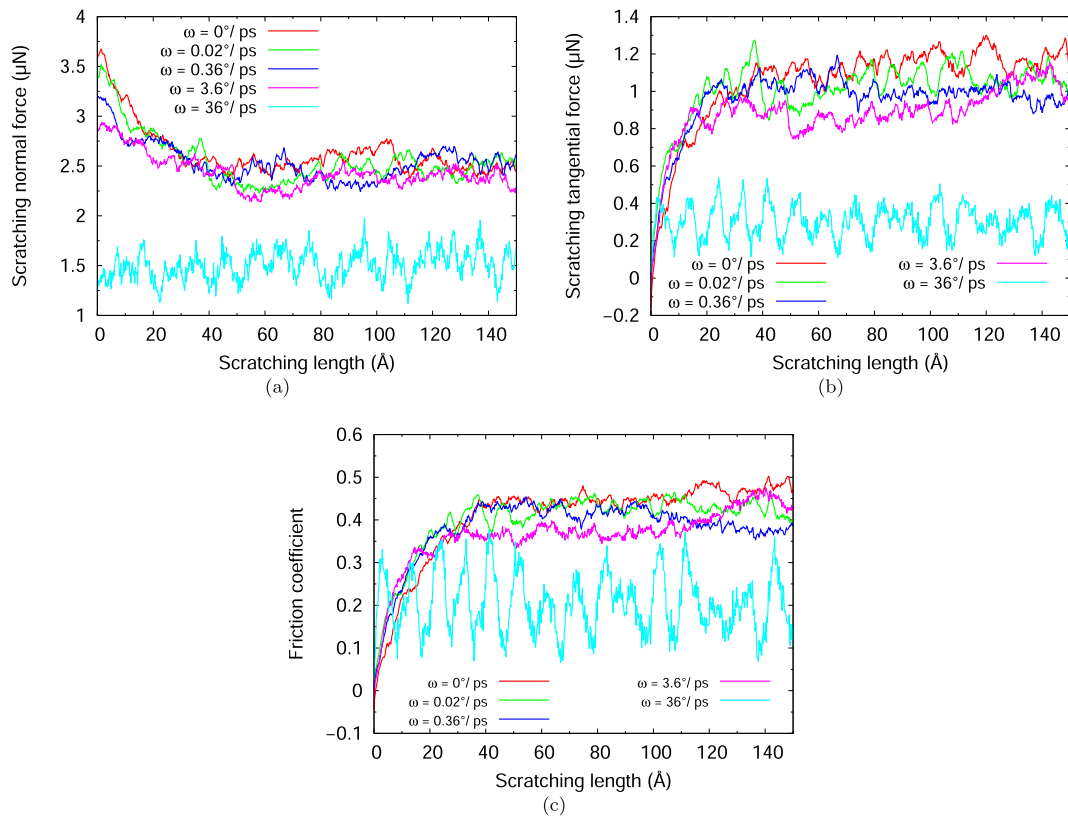


Fig. 6. As Fig. 5, but for rolling motion.

Table 2

Forces and pressures averaged over the last 100 Å of scratching after full indentation. Data are for various angular rotation velocities, ω . F_{norm} : normal force. F_{tang} : tangential force. μ : friction coefficient. H_{norm} : normal hardness. H_{tang} : tangential hardness. L_{disl} : total dislocation length after scratching and retraction of the tool.

	Drilling					Rolling				
ω ($^{\circ}$ /ps)	0	0.02	0.36	3.6	36	0.02	0.36	3.6	36	
F_{norm} (μN)	2.54	2.49	2.53	2.28	1.79	2.45	2.47	2.35	1.53	
F_{tang} (μN)	1.16	1.07	1.08	1.00	0.52	1.05	1.00	0.92	0.30	
μ	0.46	0.43	0.43	0.44	0.29	0.43	0.41	0.39	0.20	
H_{norm} (GPa)	14.91	14.60	14.64	14.02	10.12	14.24	14.54	13.91	8.42	
H_{tang} (GPa)	14.94	14.04	13.76	13.23	7.12	13.19	12.98	12.16	4.32	
L_{disl} (Å)	13058	14087	14836	14726	4258	13766	14452	14076	177	

nucleating and expanding.

4. Scratching

4.1. Forces

During scratching, two force components are measured, the normal force, F_{norm} , which is directed perpendicular to the surface, and the tangential (or lateral) force, F_{tang} , which is directed parallel to the surface, along the direction of the scratching tool. Figs. 5 and 6 show the evolution of these forces for drilling and rolling rotation, respectively. Again, as in the indentation case, the evolution is analogous for all rotation velocities except for the case of $\omega = 36^{\circ}/\text{ps}$. In the cases of lower rotation velocities, $\omega \leq 3.6^{\circ}/\text{ps}$, both normal and tangential forces feature an onset regime of around 40 Å length, corresponding to around half of the contact radius. During this early evolution, the normal force is decreased from its high value established during indentation to a smaller value necessary to keep the tool at its prescribed depth during scratching. Simultaneously, the tangential force builds up from its initially vanishing value to the force needed to move the tool laterally through the crystal.

For a quantitative comparison, we assemble the averaged values of the forces in Table 2. Both the normal and the tangential forces show a decreasing trend with ω ; at $\omega = 3.6^{\circ}/\text{ps}$, the tangential force has decreased by 20% for rolling motion and the normal force by 10% for drilling motion. A stronger influence of rolling motion on the tangential force is indeed plausible as here rotation helps the progress of the tool by moving material backwards from the front of the groove towards the rear. Similarly, drilling assists both the normal and the tangential force as it moves material both from and to the front of the groove while it destabilizes the crystalline structure of the environment. Note that the effect of rotation on the forces is already felt at quite small rotational velocities of the order of the slip-free rolling velocity, ω_{sf} .

For the highest angular velocity, $\omega = 36^{\circ}/\text{ps}$, the forces are strongly diminished by around 50%, -and even more in the case of the tangential force in rolling mode. The decrease of the forces follows the pattern described above for the smaller angular velocities in that the reduction of the tangential force in the rolling mode is particularly strong.

Strong oscillations in the forces show up for this high angular velocity; these are particularly pronounced for the rolling mode. There, as Fig. 6 shows, normal force and tangential force oscillate in anti-phase. A Fourier analysis reveals that the oscillations have a dominant

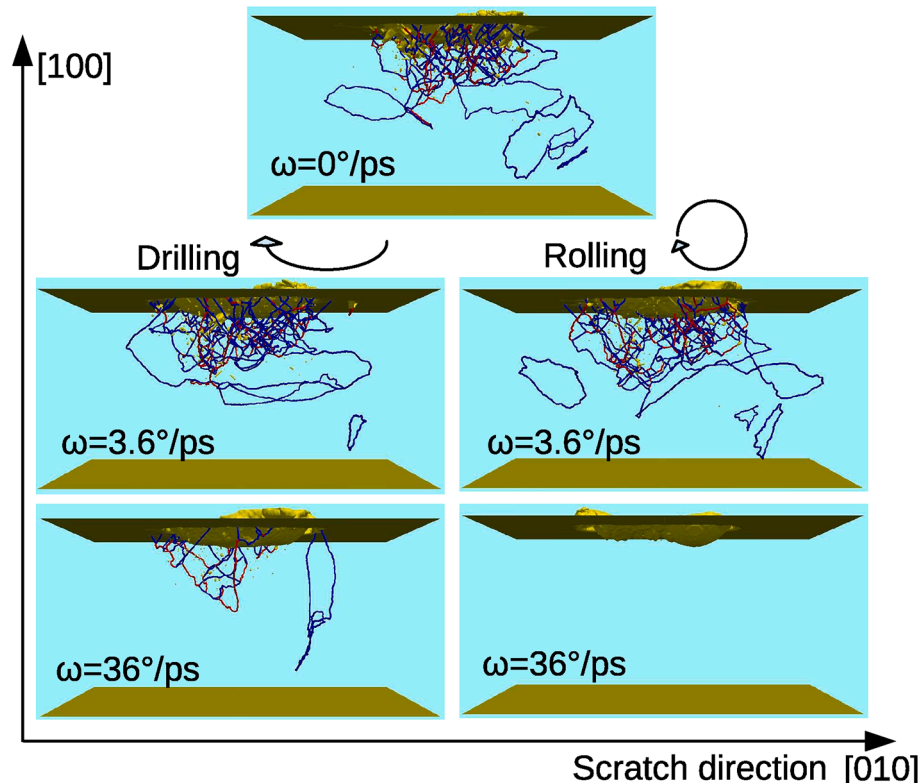


Fig. 7. Snapshots of the defects formed after scratching and retraction of the tool. Color code as in Fig. 3.

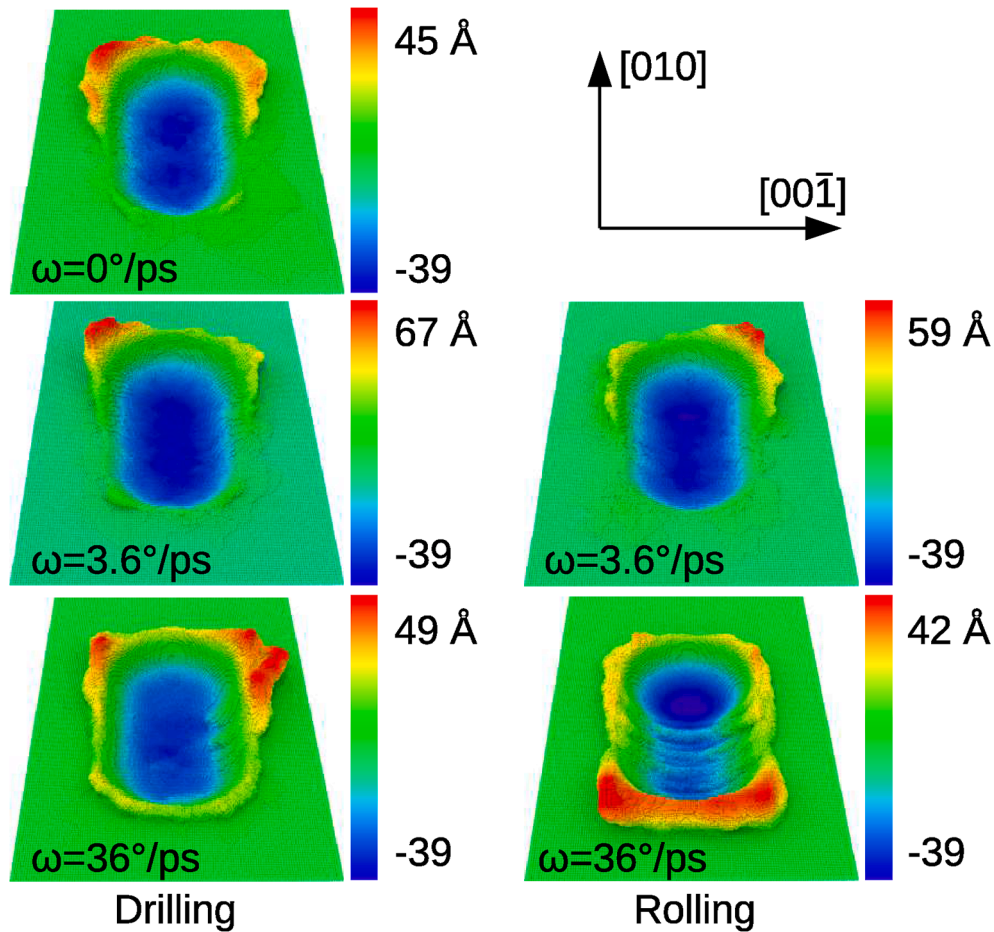


Fig. 8. Snapshots showing the scratch groove and pileup after scratching and retraction of the tool. Color denotes height above the original surface.

wavelength of 12 Å, which is considerably larger than the 2 Å that the tool needs to perform one rotation. A detailed inspection of the snapshots under rolling shows that the oscillations are accompanied by periodic variations of the amorphous zone below the tool. At the moments where F_{norm} is large and F_{tang} is small, the rear part of the amorphous zone has been reduced to a monatomic layer or even vanished. We believe that this instability is caused by the friction of the tool which transports substrate atoms from the front towards the rear of the tool. When such a transport moves atoms from the front of the tool to beneath it, the normal force needed to keep the tool at its prescribed depth increases while the tangential force is reduced. When atoms are further transported from below the tool to behind it, the normal force is decreased and the amorphous zone is allowed to partially recrystallize. Force oscillations in the drilling mode are considerably smaller, but also stem from rotational atom transport of atoms to and from the front of the tool.

The novel technique of Polyhedral Template Matching [47] promises greater reliability than Common Neighbor Analysis (CNA) in the presence of strong thermal fluctuations and strains. It uses a parameter, RMSD, to assess the matching of the local environment of atoms with the bcc crystal structure. We employ here the rather generous value of RMSD = 0.2.

The rotational velocity associated with $\omega = 36^\circ/\text{ps}$ amounts to ~ 6.3 km/s. This is slightly larger than the longitudinal speed of sound, 5.9 km/s, and clearly above the shear wave velocity, 4.54 km/s. Although supersonic dislocation motion has been observed in some cases [48], this high velocity might be connected to the impossibility of nucleating and propagating dislocations, which is replaced by amorphization to accommodate plastic deformation at that high rotation

speed. This drastic behavior change will then be similar for other metals, and occurring at the appropriate rotation velocity reaching the sound speed in the material.

The ratio of the tangential and the normal force is denoted as the friction coefficient, $\mu = F_{\text{tang}}/F_{\text{norm}}$. Table 2 shows that – as both tangential and normal force are slightly reduced by tool rotation – the friction coefficient in the drilling mode shows only little variation for the smaller rotation velocities, $\omega \leq 3.6^\circ/\text{ps}$. In the rolling mode, it monotonically decreases with ω , since rolling systematically assists the tangential motion. For the case of the highest angular velocity, $\omega = 36^\circ/\text{ps}$, the friction coefficient is sensibly reduced also in the drilling mode, as the normal force needed to keep the tool at its depth does not decrease as much as the tangential force. In the rolling mode, the anti-phase oscillations in normal and tangential force lead to large fluctuations in the coefficient of friction.

In analogy to the hardness during indentation, hardness values during scratching can be calculated. We differentiate between a normal hardness, H_{norm} , – the ratio of the average normal force to the normal area – and the tangential hardness, H_{tang} , which is obtained from the ratio of the average tangential force to the tangential area. The contact areas of the tool change dynamically during the scratching process; we take these changes into account by determining the number of Fe atoms that contact the moving tool at each time step following the atomistic methods detailed previously [23,24]. As the evolution of these contact areas does not depend much on the rotational velocity, these data are put into the SM [40]. The resulting hardness values are assembled in Table 2; we observe that the hardnesses follow the decline with rotation velocity that was discussed above for the forces.

4.2. Dislocations and scratch groove

Dislocation patterns formed under the scratch groove after scratching and retraction of the tool are depicted in Fig. 7. Analogous to the case of indentation, we refrain from showing the results for the smaller rotation velocities, $\omega < 3.6^\circ/\text{ps}$, as these are very similar to the case of $\omega = 0^\circ/\text{ps}$ shown; for the other results, we refer to the SM [40]. The dislocation pattern after scratch is characterized by a dense dislocation network of $b = \frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ dislocations filling the surroundings of the scratch groove. Some of the dislocations expand into large shear loops in the directions both in front of and behind the groove. The patterns of how these expanded loops grow are irregular and vary stochastically; no trend with ω can be observed. The total dislocation lengths assembled in Table 2 again consistently show the tendency that rotational motion increases the total dislocation length; however, the increases amount to below 14%.

The case of the highest angular velocity, $\omega = 36^\circ/\text{ps}$, is particular. As in the case of indentation, rolling motion of the tool prevents the formation of dislocations in the surrounding crystal; as discussed above, instead the groove surface is amorphized. In drilling motion, the dislocation length after indent was only around 57% of the non-rotating case, see Table 1. During scratching, L_{disl} remained fairly constant, while it increases linearly with scratch length in the case of the smaller rotation velocities. Thus the dislocation network seen in Fig. 7 for drilling at $\omega = 36^\circ/\text{ps}$ is quite small and thin; it is concentrated at the rear part of the groove, while at the front part only a single shear loop has formed.

The form of the scratch groove and the surrounding pileup is shown in Fig. 8. For a non-rotating tool, the groove has a smooth bottom at the depth of the tool indentation, $d = 40 \text{ \AA}$, and a pile-up has formed in front of the groove by the scratching process. The details of this pileup have been analyzed in detail previously [23]: The pileup does not have its maximum height at the groove front, since the $\{111\}$ slip directions responsible for atom transport above the surface move atoms not into the $[010]$ scratch direction, but at an angle of 45° to the sides. These features determine also the pileup shape for larger rotation velocities, $\omega \leq 3.6^\circ/\text{ps}$. The pile-up shown for $\omega = 3.6^\circ/\text{ps}$ displays a left-right asymmetry both in the drilling and the rolling mode; however, this is not systematic, since the rolling mode preserves the left-right symmetry of the scenario.

For the largest rotation velocity, $\omega = 36^\circ/\text{ps}$, a strong influence of rotation on both the scratch groove and the pileup is observed. For drilling rotation, the groove bottom shows only little roughness. However, groove roughness is strongly enhanced for rolling motion; here, quite regular patterns of a machining instability show up which coincides with the oscillations in the forces discussed in the context of Fig. 6 above that are caused by atom transport of the rolling tool from the front to under the tool. As a consequence, also the groove bottom is not as deep as in the other cases.

In both rolling modes, also the pileup shape is strongly distorted. For rolling motion, the pileup is concentrated at the rear of the groove rather than at its front, in agreement with the rotation direction of the tool. For drilling motion, the pileup still exhibits its concentration at the front part, but also has a large contribution spread out evenly on all sides of the groove. Here, drilling motion has transported atoms from inside the sample and deposited them on the surface at all sides of the groove rim.

5. Conclusions

In contrast to the nanomachining processes of indentation and scratching, in which a tool penetrates a surface or runs parallel to it in exclusively translational motion, the effects of rotational motion of the tool on the machining have been less studied. Such processes occur if in nanodrilling the tool rotates around an axis perpendicular to the surface; if the tool axis is parallel to the surface, such processes are termed polishing, grinding or milling. We studied here the effects of tool

rotation in a model scenario, where both the translational and rotational motion are controlled, and we could thus isolate the effects of the rotation on the machining process. By varying the rotational velocity in a wide range, we obtained the following findings.

1. For rotational speeds that are comparable to rolling without slip, the effects of tool rotation on the dislocation patterns generated, the indent pit and scratch groove, the forces during machining and the material hardness are minor. This applies both for drilling and for rolling rotation geometries.
2. At such small angular velocities, tool rotation leads to a slight decrease of the forces and the material hardness by a few percent. Tool rotation helps penetrating the material by moving atoms from the tool path. The reduction is strongest for the tangential force in rolling-mode scratching and for the normal force in drilling-mode scratching.
3. At the same rotation speed, rolling motion leads to larger effects on the machining than drilling motion. This is caused by the fact that rolling systematically shovels atoms from the front of the groove towards the rear of the groove.
4. Only for rotational speeds that strongly surpass (by an order of magnitude) the rolling-without-slip criterion, strong effects on the machining process are observed. Then dislocation plasticity is reduced and the machined material is instead prone to amorphization, which however heals after the machining process, so that the material remains defect-free. The groove develops oscillatory patterns, and the material hardness is strongly reduced.

These results were obtained for the specific case of a Fe crystal; however, they are expected to hold as well for other ductile metals.

Rotational speeds larger than that corresponding to slip-free rolling of the tool will be difficult to realize in the nanoworld. Hence our considerations of such large rotational velocities are essentially meant to provide information about the processes occurring beyond small rotation velocities.

For high rotational speeds, we find a change in deformation mechanism from dislocation slip to amorphization. Transition from slip to another deformation mode like twinning has been frequently observed as a function of strain rate [49]. However, amorphization of a metal is not easily achieved, as it quickly recrystallizes. Amorphization of nanoscale bcc metals has been achieved experimentally [50], and the slip-to-amorphization deformation transition as a function of grain size has been observed for an intermetallic [51]. Previous simulation studies of machining with a rapidly rotating tool also found sample amorphization. Wu et al. [31] studied machining of a Cu sample with a rotating tool and found that dislocation slip, which was active at zero rotation speed, vanishes for a rotation speed of $\omega = 5^\circ/\text{ps}$ (the smallest rotation velocity investigated there). For the tool radius of 1 nm used in that study, this corresponds to a circumferential tool velocity of 170 m/s; this is a smaller speed than in our case, since a smaller tool radius was used. The authors point at the similarities of amorphization under high-speed cutting (non-rotating tool) where previous studies found that dislocations annealed in the sample for high cutting speeds (180 m/s) but not at lower speeds [52].

A disordered state of matter similar to the amorphization observed here was also found by Goel et al. [35] in their study of elliptical vibration-assisted machining of pure iron. The authors conclude that due to the intensive shearing induced by the elliptical displacement of the tool, the material undergoes high disordering and plastic viscous flow. There is also some qualitative agreement between our results for a metal and some results for a covalent solid like Si, because a slip-to-amorphization transition has been observed in shocked Si [53]. In addition, a Si sample always amorphizes under machining [54,55], and the thickness of the amorphous layer increases with rotation velocity [26].

On the other hand, the results were derived for a tool of radius 10

nm, and the question arises in how far they may be generalized to tools of larger size. It is well known that plasticity in metals underlies a size effect [56–58,12] in that the hardness of the material generally decreases with tool size. Since the sample sizes can hardly be expanded for molecular dynamics simulations beyond the 20 million atoms studied here, such a question is best answered using other simulation techniques.

Another important issue concerns the interaction of the tool with the sample, which has been modeled here as purely repulsive, as it is customary in indentation and scratching studies [12,13], since these results correspond best to experimental findings. However, the question to what extent lateral friction, as it might originate from an attraction between the tool material to the sample [59], changes the results has up to now not been answered for the case of rotating tools and would be relevant to investigate in the future. Here, also the wear of the tool is a key aspect in practical applications [60,61]. The wear of diamond tools is a complex process involving micro-cracking and abrasion as well as chemical reactions and processes [62]. Molecular dynamics simulations have shown potential to provide insights into such processes [63] and might give insight into the additional wear processes occurring under tool rotation.

6. Data availability

This study is based only on data obtained using the methods described in this paper.

CRediT authorship contribution statement

Iyad Alabd Alhafez: Investigation, Software, Conceptualization, Writing - review & editing. **Carlos J. Ruestes:** Conceptualization, Writing - review & editing. **Eduardo M. Bringa:** Conceptualization, Writing - review & editing. **Herbert M. Urbassek:** Conceptualization, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Simulations were performed at the High Performance Cluster Elwertisch (RHRK, TU Kaiserslautern, Germany). IAA and HMU acknowledge support by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – project number 172116086 – SFB 926. CJR thanks support through Agencia Nacional de Promoción de la Investigación, el Desarrollo Tecnológico y la Innovación, PICT-2018-00773. EMB thanks SIIP-UNCuyo grant 06/M104.

Appendix A. Tool rotation and translation in LAMMPS

While the translational motion of the tool is straightforward to introduce in LAMMPS, the motion of the tool rotation axis during the drilling and rolling modes require more care. During the scratch process (drilling mode) and during both indentation and scratching in the rolling mode, the rotation axis points in a direction perpendicular to the translational motion of the tool. In that case, the axis of rotation needs to move at the same translational velocity as the tool. Unfortunately, LAMMPS does not provide this feature. We solve this problem by moving the entire substrate while keeping the tool rotating at a fixed position.

In order to check the validity of this procedure we perform a standard indentation simulation (non-rotating tool) with the conventional method, in which the tool moves into the sample, and compare with a simulation, in which the tool is held fixed and the sample moves up

towards the tool. As Fig. 9 demonstrates, the results are almost indistinguishable: the load drop at the onset of dislocation formation is identically reproduced, and only in the later plastic stage and in the tool retraction stage, a few minor differences show up that are caused by dislocation reactions that are subject to fluctuations. The same check can be performed for indentation in the drilling mode and also leads to good agreement (not shown). We note that in the method of moving the sample, it must be verified that not only the Newtonian atoms but also the thermostatted regions and the fixed boundary layers must move jointly (see Fig. 1).

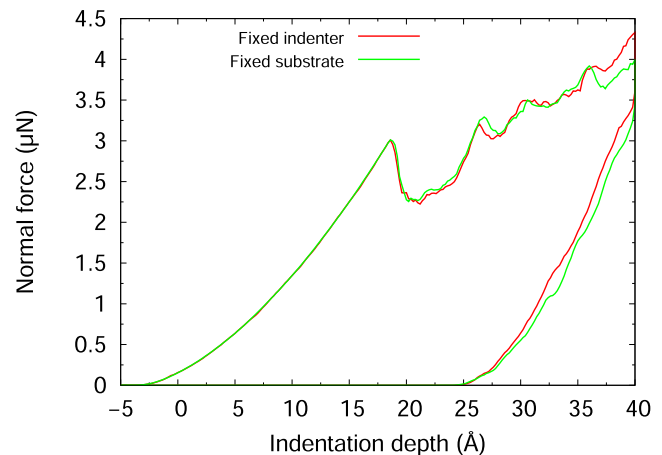


Fig. 9. Comparison of the evolution of the normal force with indentation depth for a standard simulation (fixed substrate), in which the tool moves into the sample, with a simulation (fixed indenter), in which the tool is held fixed and the sample moves into the tool.

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.commatsci.2021.110445>.

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